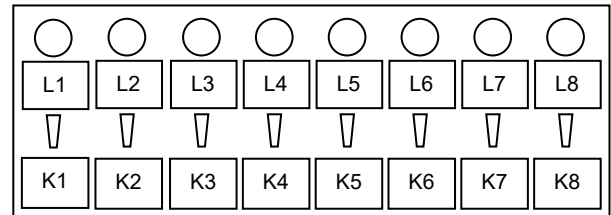


8.9.2025

Timers

Familiarize yourself with different timer delays using TIA Portal's help. How do you set the timer time? How can you see how much time the timer has counted? How should the block be connected to the program for it to be used?



- Create a function (FC1) that performs the following operations (define used TAGS first into list):
- Each logic switch (K1 - K8, I0.0 – I0.7) activates the LED (L1 - L8, Q0.0 – Q0.7) above it.
 - Add a function to switch K1: When switch K1 is moved to position 1, the LED above it will light up after 10 seconds if the switch is left on.
 - Add function K2 to the switch: When switch K2 is moved to position 1, LED L2 lights up immediately. It stays on as long as the switch is on and for an additional 5 seconds afterwards.
 - Add function K3 to the switch: When switch K3 is moved to position 1, the LED above will light up for 10 seconds, if the switch is left on.
 - Add function to switch K7: When switch K7 is moved to position 1, the LED above it will light up for 10 seconds, even if the switch is not left in position 1.
 - Add function K8 to the switch: When switch K8 is moved to position 1, the LED above it will light up after 10 seconds, even if the switch has only been used in the ON position.
- Build an appropriate reset for the timer.

Implementation and simulation:

Make the application in suitably sized parts. Once you have completed a certain whole, test its functionality by simulating.

After this exercise, you should be familiar with the basic types of timers, know their differences, be able to choose a suitable timer for each need, and understand the principles of how timers work.

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